

GROUP 4

Assignment 4 Report: Representational Similarity Between Auditory Models and Human STG

1. Introduction

The aim of our assignment was to study whether artificial auditory models develop internal representations that resemble neural representations in the human superior temporal gyrus (STG), and the ways in which this resemblance depends on model depth, training, and architecture. The responses were taken from the Santoro dataset, containing STG responses to 288 sounds. These responses were compared with representations extracted from four local neural-network architectures—waveform, uninspired, inspired, and a custom CorticalCRNN—as well as the pretrained YAMNet model.

The CorticalCRNN was designed as a simplified, biologically inspired model of hierarchical auditory processing. Its convolutional stages—core, belt, and parabelt—are motivated by the organization of primate auditory cortex described by Kaas and Hackett (2000), in which auditory information progresses from primary-like core regions through belt regions to higher-level parabelt areas. The recurrent STG-like stage was added to integrate information over time. The overall convolutional-recurrent architecture was inspired by Çakır et al. (2017), who showed that convolutional layers can extract local time-frequency features while recurrent layers capture longer-term temporal information in sound. The model should therefore be viewed as a loose computational analogy of auditory processing rather than a direct simulation of the brain. We conducted 5 training runs of the model, 20 epochs each.

Conducted analysis was primarily contrastive: trained models were compared with untrained versions, different architectures were compared with one another, and early versus late layers were contrasted. It also contained a constructive component because the CorticalCRNN was designed specifically as a biologically motivated auditory model and then tested against brain data.

For all four architectures we had five independently trained runs, and five reproducible untrained initializations were generated for comparison. YAMNet was treated as one fixed pretrained reference. Detailed layer-level outputs, confidence intervals, additional tables and more detailed analysis are provided in the notebook markdown cells.

For more detailed information we refer to the notebook markdown.

2. Methods

2.1. Representational similarity analysis

Because the brain responses and model activations have different numbers of features, they cannot be compared directly. Instead, we used Representational Similarity Analysis (RSA). For this, each brain or model representation was converted into a Representational Dissimilarity Matrix (RDM).

For the STG data and for every model layer, each of the 288 sounds was represented by its pattern of activity. We then calculated the correlation distance between every pair of sounds. This produced a 288×288 RDM. Small values mean that two sounds produced similar representations, while large values mean that their representations were more different. Because the matrix is symmetric, only the unique pairwise values from the upper half of the matrix were used for comparison.

Each model RDM was compared with the STG RDM using three measures: Pearson correlation, Spearman correlation, and Kendall's tau-b. Spearman and Kendall mainly test whether the model and brain rank sound pairs in a similar order from most similar to most different. Pearson is more sensitive to the actual size of the dissimilarities.

For the trained models, activations were analysed separately for each training run and compared with an untrained version of the same architecture. The effect of training was calculated as trained RSA minus untrained RSA. A positive value therefore means that training on ESC-50 made the model's representations more similar to the STG representations.

2.2. Multiple runs and statistical treatment

Training a neural network involves randomness, so two networks with the same architecture can learn slightly different representations. For this reason, five independent training runs were analysed for each model. Statistics were first calculated within each run and only then averaged across runs. This avoids treating different layers of the same network as if they were independent observations.

For the main results, we report the mean together with a 95% confidence interval across the five runs. The same approach was used for overall brain alignment, the best-aligned layer, the effect of training, the relationship between layer depth and brain alignment, and the category-separation analysis. YAMNet was available only as one pretrained model, so no confidence interval across runs could be calculated for it.

We did not perform a formal statistical test to decide whether one architecture was significantly better than another. The model comparisons are therefore descriptive and should be interpreted based on the size of the differences and the amount of variation across runs.

Because the models contain different numbers of layers, layer depth was rescaled from 0 for the first layer to 1 for the last layer. For each run, we then measured whether brain alignment tended to increase or decrease with model depth using Spearman correlation.

We also measured how clearly sounds with the same sound-source label, such as bird, dog, cat, or monkey, were grouped together. This was done using the silhouette score based on the representational distances.

Finally, t-SNE was used to visualize how the representations were organized. Since separate t-SNE maps cannot be meaningfully averaged, only one representative run, run 0, is shown for each model. These plots are therefore used only as visual examples and not as statistical evidence.

3. Results

3.1. Overall model alignment with STG

Looking across entire model architectures, the trained inspired and CorticalCRNN models show the best activity match the superior temporal gyrus (STG) among our locally trained models, while the waveform model consistently showed negative alignment with the brain. Across all three metrics, average alignment scores were small but consistent, indicating a modest, real similarity to STG rather than a close reproduction. Specifically, Pearson mean alignment was 0.0179 for inspired, 0.0159 for CorticalCRNN, 0.0086 for un-inspired, and -0.0428 for waveform. Spearman scores followed the same general pattern at 0.0160, 0.0125, 0.0056, and -0.0387 , respectively, while Kendall metrics produced comparable results.

Looking at local peak layers revealed stronger individual alignment. CorticalCRNN peaked most often at its parabelt_conv layer, reaching a mean peak Spearman RSA of 0.0509. The inspired model usually peaked near its classifier with a mean peak Spearman RSA of 0.0576. The un-inspired model peaked around fc_layers.1 at 0.0330, while the waveform model stayed negative even at its peak around -0.0190 . YAMNet displayed higher individual peaks, reaching a Spearman RSA of 0.098 at layer12relu, a Pearson of 0.0766 at layer08relu, and a Kendall of 0.0633 at layer12relu.

3.2. Effect of training

Training effects depended significantly on the specific layer. Across architectures, the clearest positive gains in STG similarity occurred in intermediate layers rather than across the entire network in an uniform way..

For CorticalCRNN, Spearman RSA increased by 0.0486 at belt_conv (with a 95% confidence interval of [0.0151, 0.0821]) and by 0.0620 at parabelt_conv [0.0204, 0.1036], with Pearson and Kendall showing the exact same pattern. In contrast, the stg_rnn stage had a negative mean training effect, and the classifier showed little consistent improvement. The inspired model behaved similarly, as its second and third convolutional layers improved substantially after training, gaining 0.0622 and 0.0725 in Spearman correlation, respectively, while its recurrent layer slightly declined. The un-inspired model also showed positive training effects in its convolutional layers, but negative effects in later fully connected and classifier stages. Waveform improved relative to its untrained state in several middle hidden layers, but its overall RSA remained negative. So training made waveform less dissimilar from STG without giving it positive overall alignment.

3.3. Model depth

Depth had a strong positive relationship with STG alignment for inspired and uninspired models. For trained inspired, the mean Spearman depth correlation was 0.90, and for trained uninspired it was 0.886.

CorticalCRNN differed significantly. Its trained Spearman depth correlation was only 0.40 [0.103, 0.697] because alignment increased through the early convolutional layers but then declined after parabelt_conv, with Pearson showing an even weaker relationship overall. This shows that in inspired and uninspired models, deeper processing predicted in general stronger brain alignment, but in CorticalCRNN the specific computation a layer performed mattered far more than simply how deep it was.

Some untrained models already showed strong positive depth gradients. Untrained CorticalCRNN had a Spearman depth correlation of 0.72, and uninspired reached 0.92. This shows that network architecture alone creates depth-dependent structure, and training simply modifies that structure rather than creating it from scratch.

3.4. Category structure

The STG silhouette score was -0.202 , indicating that sound categories from the ESC-50 dataset do not form compact, clearly separated clusters in actual human brain activity.

For trained CorticalCRNN, silhouette scores were -0.120 , -0.127 , and -0.162 across its three convolutional stages, but dropped sharply to -0.357 at stg_rnn and -0.488 at the classifier. Inspired showed a similar pattern, while uninspired dropped significantly in fully connected layers. Waveform produced very negative silhouette scores throughout training, reaching approximately -0.79 at its classifier.

Category separability was also negatively linked to STG alignment in most models. For trained inspired, the correlation was around -0.82 across all RSA variants. Trained uninspired also showed a strong negative relationship, and trained CorticalCRNN showed a weaker negative rank-based association of about -0.36 . Thus, representations that organize ESC-50 categories into cleaner clusters are not necessarily more brain-like.

3.5. Visual patterns (t-SNE)

The t-SNE plots generally supported the statistical results. The actual STG brain responses were broadly mixed, with sounds from different categories overlapping rather than forming clearly separated groups. In the trained Inspired and CorticalCRNN models, the representations became more organized across the layers, especially in the middle convolutional stages. However, clear-looking groups or branches in the later recurrent and classification layers did not necessarily mean that the categories were better separated or that the model had become more similar to the brain. The t-SNE plots are therefore useful for showing how the model's representation changes during processing, but they cannot by themselves tell us whether a representation is brain-like.

4. Discussion

The main conclusion is that training does not make an auditory network uniformly more brain-like. Instead, ESC-50 training mainly improves intermediate feature representations. In both inspired and CorticalCRNN, convolutional layers benefited strongly, whereas recurrent or classification stages often lost part of that improvement.

In the custom CorticalCRNN, the biologically motivated convolutional hierarchy seems to perform well relative to the model's other stages. `parabelt_conv` was the modal peak layer and showed one of the strongest positive effects of training.

In contrast, the `stg_rnn` did not become more STG-like after training. Calling the layer STG-inspired therefore does not guarantee neural similarity. The GRU may compress temporal information too aggressively, or ESC-50 training may encourage it to preserve task-relevant information that differs from the structure measured in STG.

The inspired model produced slightly higher architecture-level alignment than CorticalCRNN, whereas waveform performed substantially worse. One possible explanation is representational format: convolutional processing of spectrotemporal input may impose inductive biases closer to auditory cortical processing than a waveform-oriented architecture. Nevertheless, the relatively small RSA values show that none of the architectures provides a close model of STG.

The category analysis showed that stronger brain alignment did not necessarily coincide with cleaner separation of the Santoro sound-source labels. This is plausible because STG itself did not show especially strong clustering according to these discrete object or sound-source categories. Real cortical representations likely preserve several dimensions at the same time—including acoustic similarity, temporal structure, source identity, and broader semantic relationships—rather than organizing every sound primarily around a single categorical label.

The t-SNE analysis is therefore best interpreted as a complementary visualization. It illustrates how training reorganizes the geometry of the model representations and can help reveal transitions between processing stages, but visually separated clusters should not be interpreted as direct evidence of stronger sound-source categorization or better brain alignment.

5. Future work

The current CorticalCRNN captures only a small subset of auditory neuroanatomy. Based on the lecture and assignment themes of hierarchy, recurrence, tonotopy, and connectivity, a logical next step would be to make the architecture less strictly serial.

Rather than connecting `core` → `belt` → `parabelt` → STG in a simple chain, future models could use atlas-informed connectivity with parallel and reciprocal connections between auditory regions. Recurrent feedback from higher to lower stages could be introduced instead of restricting recurrence to a single GRU. Multiple recurrent timescales may also be useful, allowing early regions to integrate short acoustic windows while higher regions integrate information over longer periods.

Moreover, pretraining on a larger auditory dataset, using self-supervised prediction or contrastive learning before ESC-50 fine-tuning, could lead to more general/less specialized auditory representations.

6. References

Kaas, J. H., & Hackett, T. A. (2000). Subdivisions of auditory cortex and processing streams in primates. *Proceedings of the National Academy of Sciences of the United States of America*, 97(22), 11793–11799. doi:10.1073/pnas.97.22.11793.

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